Auteur: Siebe Haché
Studierichting: Informatica
Oefeningenreeks 2B nr. 6.9

Toon aan dat $5\,\mathrm{Bgtan}\,\frac{1}{7} + 2\,\mathrm{Bgtan}\,\frac{3}{79} = \frac{\pi}{4}$.

$\alpha = \mathrm{Bgtan}\,\frac{1}{7}$ en $\beta = \mathrm{Bgtan}\,\frac{3}{79} \Rightarrow \tan\alpha = \frac{1}{7}$ en $\tan\beta = \frac{3}{79}$

We gebruiken: $\tan 2\theta = \dfrac{2\tan\theta}{1-\tan^2\theta}$

$$\tan 2\alpha = \frac{2\cdot\frac{1}{7}}{1-\left(\frac{1}{7}\right)^2}$$

$$= \frac{\frac{2}{7}}{1-\frac{1}{49}} = \frac{\frac{2}{7}}{\frac{48}{49}}$$

$$= \frac{7}{24}$$

Somformule geeft:

$$\tan 3\alpha = \tan(2\alpha+\alpha)$$

$$= \frac{\frac{7}{24}+\frac{1}{7}}{1-\frac{7}{24}\cdot\frac{1}{7}}$$

$$= \frac{\frac{73}{168}}{\frac{23}{24}} = \frac{73}{161}$$

$$\tan 4\alpha = \tan(3\alpha+\alpha)$$

$$= \frac{\frac{73}{161}+\frac{1}{7}}{1-\frac{73}{161}\cdot\frac{1}{7}}$$

$$= \frac{\frac{672}{1127}}{\frac{1054}{1127}} = \frac{336}{527}$$

$$\tan 5\alpha = \tan(4\alpha+\alpha)$$

$$= \frac{\frac{336}{527}+\frac{1}{7}}{1-\frac{336}{527}\cdot\frac{1}{7}}$$

$$= \frac{2879}{3353}$$

Nu $\tan 2\beta$:

$$\tan 2\beta = \frac{2 \cdot \frac{3}{79}}{1 - \left(\frac{3}{79}\right)^2}$$

$$= \frac{\frac{6}{79}}{1 - \frac{9}{6241}} = \frac{\frac{6}{79}}{\frac{6232}{6241}}$$

$$= \frac{237}{3116}$$

We bepalen $\tan(5\alpha + 2\beta)$.

Somformule geeft:

$$\tan(5\alpha + 2\beta) = \frac{\frac{2879}{3353} + \frac{237}{3116}}{1 - \frac{2879}{3353} \cdot \frac{237}{3116}}$$

$$= \frac{\frac{9765625}{10447948}}{\frac{9765625}{10447948}}$$

$$= 1$$

Dus

$$5\alpha + 2\beta = \mathrm{Bgtan}(1) = \frac{\pi}{4}$$

$$\Rightarrow 5\,\mathrm{Bgtan}\,\frac{1}{7} + 2\,\mathrm{Bgtan}\,\frac{3}{79} = \frac{\pi}{4}$$

References